

From the baseline to state-dependent error bounds

A step-by-step record of the work completed so far, with the mathematical derivation and a guide to the code. Prepared for Satyam Awasthi. Updated 11 September 2026.

Current result

We recovered the original uncalibrated observer radius from its LMI and constructed a smaller local process envelope using the same metric and gains. This is supported by numerical audits; continuous-domain and full MPC guarantees remain open.

The work in five steps

Step	What was done	Where explained
1. Reproduce	Initialized public dependencies and ran the two offline SDPs.	Page 3
2. Understand	Separated true-to-estimate and estimate-to-nominal errors.	Pages 2 and 4
3. Prototype	Tried dynamic norm bounds; found extra conservatism.	Page 4
4. Recover	Derived squared-radius dynamics directly from the original LMI.	Page 5
5. Localize	Reduced the disturbance envelope and compared tube propagation.	Pages 6-8

What this report is about

The focus is the offline observer/controller certificates and error-tube propagation. We have not yet run a closed-loop quadrotor simulation with the new tubes, integrated them into online MPC, or recomputed UQ for a physical system.

How to read it

Read pages 2-3 for the model and baseline. Pages 4-7 derive the new bounds. Page 8 explains the results, page 9 separates evidence from guarantees, and page 10 maps the equations to scripts.

Source basis: the ROHMPC paper (arXiv:2508.07045v1), the pinned public mpc-sdp implementation, and our saved MATLAB runs. Precise source revisions and file locations are on page 10.

The model and the two errors

The observer, nominal prediction and true plant are different trajectories. Keeping them separate explains the two tubes.

$$\dot{x} = f_0(x, u) + E_0 D(a(v)) w^0, \quad y = Cx + F\eta \quad (1)$$

$$\dot{\hat{x}} = f_0(\hat{x}, u) + L(y - C\hat{x}), \quad \dot{z} = f_0(z, u_{\text{nom}}) \quad (2)$$

$$u = u_{\text{nom}} + K(\hat{x} - z) \quad (3)$$

Here f_0 includes the fixed modeled disturbance bias. The residual disturbance w^0 lies in the original centered box. D scales only acceleration channels; all other process channels and measurement-noise bounds are unchanged.

Symbol	Meaning
$x, \hat{x}, z$	True state, observer estimate, nominal trajectory.
$e = x - \hat{x}$	Estimation error: what the observer does not know.
$\delta = \hat{x} - z$	Tracking error of the estimate relative to the nominal trajectory.
P	Positive-definite metric used to measure both errors.
$r(t), s(t)$	Upper bounds on estimation and tracking errors in that metric.
$R(t) = r(t)^2$	Squared estimation radius, used in the improved derivation.
T	Cholesky factor: $T^T T = P$. Not the scalar R .

$$\|e\|_{P^\delta} \leq r, \quad \|\delta\|_{P^\delta} \leq s, \quad \|x - z\|_{P^\delta} \leq r + s \quad (4)$$

A radius is a bound, not an observed error

These are weighted state-space radii, not distances in meters. A smaller computed radius does not by itself mean the actual estimation error is smaller.

What the baseline run computed

An LMI is a matrix inequality. An SDP is an optimization problem containing semidefinite constraints, typically many LMIs.

$$M(q) = M_0 + \sum_i q_i M_i \preceq 0$$

(5)

The inequality means every eigenvalue is nonpositive, or equivalently $a^T M(q) a$ is nonpositive for every a . It is not an entrywise inequality. YALMIP constructs the optimization problem; MOSEK solves it.

Offline problem	Computed ingredients
Robust-design SDP	X, Y , metric $P\text{-}\delta = X^{-1}$, gain $K = YX^{-1}$, tightening coefficients, observer radius ϵ and auxiliary envelopes. The observer gain L is prescribed in this run.
Terminal-cost SDP	A separate terminal-cost matrix P . This is different from the error metric $P\text{-}\delta$.

The actual 1,024-point operating grid

Quantity	Values used
Four rotor commands	Each: 1.3935627 and 1.6335627 (model input units).
Roll, pitch, yaw	Each: -0.1 and +0.1 rad.
Three angular rates	Each: -0.3 and +0.3 rad/s.
Position and linear velocity	All fixed to zero in the grid.

The count is $2^4 \times 2^3 \times 2^3 = 1,024$. Nominal drag is zero, so these Jacobians are independent of linear velocity; this is not a hover-only controller. The permitted velocity box remains $[-2, 2]^3$ m/s. Construction and verification use the same grid settings.

Reproduction outcome

Both SDPs returned OPTIMAL. The repeated run took 265 seconds and reproduced objectives 1494.858118 and 38.564610. The uncalibrated observer bound was $\epsilon = 0.07442638619$. Several tightened constraint margins were negative: successful offline optimization did not yield usable uncalibrated MPC constraint sets.

The repository subsequently uses empirical calibration to reduce conservatism. That stage was not used in our bound comparisons.

Where the propagation equations come from

Subtract the observer from the plant to obtain estimation error. Subtract the nominal predictor from the observer to obtain tracking error.

$$\dot{e} = f_0(x, u) - f_0(\hat{x}, u) - LCe + E_0 D(a) w^0 - LF\eta \quad (6)$$

$$\dot{\delta} = f_0(\hat{x}, u) - f_0(z, u_{\text{nom}}) + LCe + LF\eta \quad (7)$$

Process disturbance enters estimation error directly. It affects estimate-to-nominal tracking through LCe. Measurement noise also enters tracking directly through the observer correction. The plant-to-nominal error x-z contains both errors.

First prototype: separate norm bounds

$$\dot{r} = -\rho_o r + b_{\text{other}} + a_{\text{upper}} b_{\text{acc}} + b_{\text{noise}} \quad (8)$$

$$\dot{s} = -\rho_c s + \gamma r + b_{\text{noise}} \quad (9)$$

Contraction removes existing error. The b terms bound metric-norm contributions: other process disturbances, acceleration disturbances, and observer-injected measurement noise. The scalar gamma bounds the gain from estimation error into tracking error.

$$b_{\text{noise}} = \max_{\eta} \|TLF\eta\|_2, \quad \gamma = \|TLCT^{-1}\|_2 \quad (10)$$

Separate maximizations followed by the triangle inequality made this first construction conservative. It reused the baseline metric and gains, but selected contraction rates from numerical audits with a margin.

First prototype comparison	Final estimation radius
Global forcing in the new norm formula	0.105718
Local forcing in the same norm formula	0.097925
Original uncalibrated SDP bound	0.074426

Why we changed the derivation

Local information helped within the first prototype, but it did not beat the original SDP bound. We therefore returned to the original quadratic observer LMI instead of summing separate norm bounds.

Recover the original observer bound

This is the key bridge from the authors' offline certificate to a dynamic squared estimation radius.

Write $X = (P\text{-}\delta)^{-1}$, $A_o = A - LC$, $\lambda = \lambda_{\epsilon}$, $W = W_{\text{bar}}$ and $H = H1_{\text{bar}}$. The metric is constant, so $\dot{X}$ is zero. Define the symmetric disturbance block:

$$S(b) = \begin{bmatrix} 0 & b \\ b^T & 0 \end{bmatrix}$$

$$S(E_0 w^0) \leq W, \quad S(-LF\eta) \leq H \quad (11)$$

$$W + H + \text{diag}(XA_o^T + A_o X + \lambda X, -q_0) \leq 0 \quad (12)$$

$$q_0 = \lambda \epsilon^2 \quad (13)$$

Adding these inequalities cancels the auxiliary matrices. Evaluate the resulting matrix quadratic form at $[P\text{-}\delta e; 1]$. With $V = e^T P\text{-}\delta e$, this gives:

$$\dot{V} \leq -\lambda V + q_0 \quad (14)$$

For nonlinear dynamics, A must cover the Jacobians along the segment from the estimate to the true state. The algebra is exact if the underlying matrix inequalities hold on that domain; our present checks are numerical and finite.

Propagate the bound rather than fixing it

$$\dot{R} = -\lambda R + q_0, \quad R(0) \geq V(0), \quad r = \sqrt{R} \quad (15)$$

$$R(t) = \epsilon^2 + [R(0) - \epsilon^2]e^{-\lambda t} \quad (16)$$

Exact recovery of the baseline formula

If $R(0) = \epsilon^2$, then $r(t) = \epsilon$ at every time. Numerically, $\sqrt{q_0/\lambda} = 0.07442638619$, exactly matching the saved baseline. A zero initial radius is valid only when initial estimation error is zero.

Reduce the process envelope locally

We keep the metric, gains and noise envelope. Only the process-envelope allowance changes with the prescribed uncertainty scale.

$$W = \begin{bmatrix} Q & h \\ h^\top & c \end{bmatrix}$$

In the saved design Q is positive definite. The Schur complement converts the process LMI into a quadratic bound:

$$S(E_0 D(a) w^0) \leq W \iff (E_0 D(a) w^0 - h)^\top Q^{-1} (E_0 D(a) w^0 - h) \leq c \quad (17)$$

$$g(a) = \max_{w^0 \in \text{vert}(\mathcal{W}_0)} (E_0 D(a) w^0 - h)^\top Q^{-1} (E_0 D(a) w^0 - h) \quad (18)$$

The quadratic is convex in disturbance, so its maximum over the box occurs at a vertex. Each vertex expression is convex in a ; therefore $g(a)$ is convex. Because the centered sets are nested, $g(\beta)$ is no larger than $g(1)$.

$$g(a) \leq g(1) - \frac{1-a}{1-\beta} [g(1) - g(\beta)] \quad (19)$$

We use 99% of this available reduction, retaining a numerical cushion:

$$\Delta = 0.99[g(1) - g(\beta)], \quad c_{\text{local}}(a) = c - \Delta \frac{1-a}{1-\beta} \quad (20)$$

$$q(a) = q_0 - \Delta \frac{1-a}{1-\beta} \quad (21)$$

Keep Q and h fixed and replace c by c_{local} . Decreasing the process matrix's bottom-right entry and decreasing q by the same amount leaves equation (12) unchanged. The new process envelope still contains the local disturbance set, conditional on the original inequalities.

Why this avoids the shared-envelope bottleneck

There is now a family $W_{\text{local}}(a)$, rather than one worst-case process allowance at every state. At $a = 1$ it recovers the original W and q_0 . No new SDP was needed for this particular nested-box construction.

Use a bound on the unknown true speed

The disturbance depends on true velocity, but the true state is not known. Both error tubes must therefore enter the local uncertainty calculation.

$$a(v) = \beta + (1 - \beta) \frac{\|v\|_2^2}{v_{\max}^2}, \quad \beta = 0.25, \quad v_{\max} = \sqrt{12} \text{ m/s} \quad (22)$$

Only acceleration-disturbance amplitudes use this scale: 25% of the original maximum at zero speed and 100% at the domain's maximum speed. Other process disturbances and measurement noise remain at their baseline bounds.

$$c_v = \|S_v T^{-1}\|_2, \quad \|v_{\text{true}}\|_2 \leq \|v_z\|_2 + c_v(\sqrt{R} + s) \quad (23)$$

$$v_{\text{upper}} = \min \{v_{\max}, \|v_z\|_2 + c_v(\sqrt{R} + s)\} \quad (24)$$

$$a_{\text{upper}} = \beta + (1 - \beta)(v_{\text{upper}}/v_{\max})^2 \quad (25)$$

The improved coupled propagation

$$\dot{R} = -\lambda R + q(a_{\text{upper}}), \quad r = \sqrt{R} \quad (26)$$

$$\dot{s} = -\rho_c s + \gamma \sqrt{R} + b_{\text{noise}} \quad (27)$$

R affects s through the observer correction. Conversely, both radii affect the upper bound on true speed, which sets the process forcing in R. A low-speed region permits contraction over time; it does not justify an instantaneous reduction of estimation error.

Assumptions required for the comparison

Initial radii must contain initial errors. States, estimates, nominal states and relevant inputs must remain in the domain supporting the LMIs. The speed cap is valid only under that domain assumption. These propagation equations do not themselves prove domain preservation.

The scalar forcing is nondecreasing with uncertainty scale. With the same initial radii and valid derivative inequalities, the local coupled comparison can be ordered below the corresponding global-forcing comparison. Strict improvement is trajectory-dependent, not universal.

What the numerical comparison shows

The improved LMI-based construction uses the same metric, gains, noise bounds and initial radii in both comparisons: $R(0) = \epsilon^2$ and $s(0) = 0$.

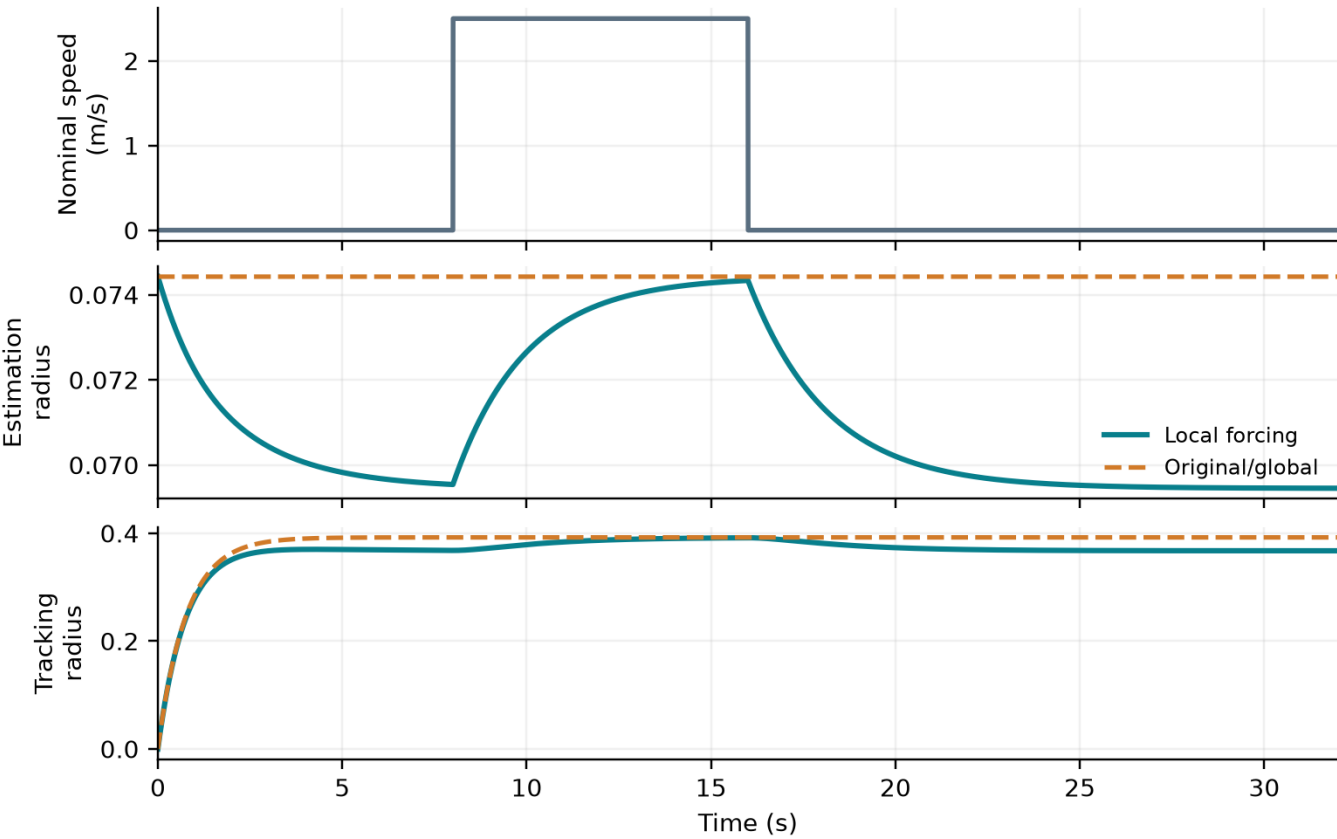


Figure 1. Prescribed nominal-speed schedule and resulting comparison-ODE radii. This is not a simulated quadrotor trajectory and need not be dynamically feasible.

At t = 32 s	Original/global	Local forcing
Estimation radius	0.07442638619	0.06945456129
Tracking radius	0.392384	0.367532

Interpretation

The final estimation radius is 6.68% smaller in this example. This improves on the original uncalibrated bound within the baseline-equivalent propagation comparison. It does not establish lower actual estimation error, universal improvement, or closed-loop MPC safety.

Parameters: λ approximately 0.497029; $q_0 = 0.002753187883$; Δ approximately 0.000441487. The first prototype's larger radius is not used as the baseline here.

Evidence, assumptions and unfinished work

Distinguish an algebraic implication, a numerical check and a closed-loop robustness claim.

Evidence completed	What it establishes
Baseline recovery	The global squared-radius formula reproduces the original epsilon algebraically and numerically.
Local-envelope derivation	Convexity and the Schur complement justify the local reduction, conditional on the original matrix inequalities.
Numerical LMI audits	Original process/noise vertices and 1,024 dynamics points checked against the saved tolerance.
Local tests	All 4,096 process vertices at 31 scales; 1,968 admissible nonlinear error configurations; scalar ODE and trajectory-ordering tests.

The maximum original matrix residuals were approximately $+5.6 \times 10^{-8}$, within the code's tolerance of 10^{-6} . They were not strictly negative. A formal certificate must also address numerical residuals, not merely accept the solver status.

The required next steps

Priority	Work still required
1. Domain coverage	Bound nonlinear Jacobians between operating points and verify observer/tracking inequalities over that enclosure.
2. Numerical rigor	Treat residuals and conditioning explicitly; introduce verified margins or certified numerical bounds.
3. MPC construction	Adapt initialization, constraint tightening, terminal conditions and recursive-feasibility arguments.
4. Closed-loop evaluation	Use our simulator to check actual errors, constraints and tube containment through low/high-uncertainty transitions.

Uncertainty assumptions are separate from controller guarantees

Our local sets are prescribed subsets of the original UQ envelope. Containment in a global set does not prove real disturbances belong to the smaller local sets. Applying the claim to Gazebo or hardware requires additional model/data validation.

Code map and reproducibility

Use the WSL checkout: it preserves the distinct filenames `compute_L.m` and `compute_I.m`. Windows MATLAB, CasADi, YALMIP and MOSEK were used for the baseline.

Original offline design: read in this order

File in mpc-sdp	Role
main.m	Coordinates both SDPs and saves <code>offline_design.mat</code> . It overwrites that output.
FalconModelT.m	Dynamics, Jacobians, constraints, grid and uncertainty vertices.
compute_Pdelta_Kdelta_cs_co.m	Main robust-design SDP and post-solve checks.
compute_check_epsilon_RPI_LMIs.m	Observer envelope inequalities (11)-(12).
compute_w_o_bar_c.m	Tracking growth $\gamma \times \epsilon + \text{noise}$.
compute_P.m	Terminal-cost SDP; separate from the error metric.

Our research files

File	Role
run_lmi_radius_recovery.m	Current experiment, including local <code>lmi_rhs</code> and validation checks.
LMI_RADIUS_DERIVATION.md	Detailed derivation and scope of numerical evidence.
run_dynamic_bound_demo.m	Earlier, more conservative norm-based prototype.
bound_rhs.m; check_error_dynamics.m	RHS and nonlinear tests for the earlier prototype only.

Project root: `/home/satyam/ROHMP-State-Dependent-Uncertainty`
Original MATLAB folder: `src/catkin_ws/src/mpc/mpc_solver/scripts/include/offline_computations/mpc-sdp`
Our code: `research/state_dependent_bounds`
Saved baseline: `work/offline-sanity-20260910/offline_design.mat`
Improved-run evidence: `research/state_dependent_bounds/results/lmi_20260910_222354`

Sources and version anchors

Benders et al., *From Data to Safe Mobile Robot Navigation: An Efficient and Modular Robust MPC Design Pipeline*, Sections II-IV, especially (7e), (12)-(13). arxiv.org/abs/2508.07045
Public code: github.com/dbenders1/rohmpc. Pinned `mpc-sdp` revision: `ca4ba0c7b6f27bd2867b39a5bf5d666887901ecf`. Input-data submodule revision: `5f3397db1e170496cc8c2b72977b8580292426c0`. Results and derivations in this report are from our local runs; they are not additional claims made by the original paper.